

Content Delivery, Ingestion and Data Transfer

Moving data in, out and across distance.

DOMAIN 3 of 4 EXAM WEIGHT 24% MODULE 9 of 22

Latency is mostly **distance**. Every service in this module either shortens the distance or removes the round trip.

CloudFront, and what it is actually for

- ◆ **Caches at edge locations** worldwide, so users fetch from nearby rather than from the origin Region.
- ◆ **Works for dynamic content too**, not just static files. It still shortens the network path.
- ◆ **Origin Access Control** lets CloudFront read a private S3 bucket, so the bucket never needs to be public.
- ◆ **Signed URLs and signed cookies** restrict access to specific users for a limited time.
- ◆ **Reduces origin cost**, since cached requests never reach S3 or the load balancer.

ON THE EXAM

'Users worldwide report slow load times' is CloudFront. 'Serve S3 content without making the bucket public' is CloudFront with Origin Access Control.

CloudFront or Global Accelerator?

CLOUDFRONT

Caches content at the edge

- HTTP and HTTPS only
- Stores a copy near the user
- Ideal for static and cacheable content
- Charged by data transfer out

GLOBAL ACCELERATOR

Improves the network path

- TCP and UDP, any protocol
- Caches nothing at all
- Ideal for APIs, gaming, VoIP, non-HTTP
- Two static anycast IP addresses

THE TIEBREAKER

Cacheable content means CloudFront. Non-HTTP traffic, static IPs, or fast regional failover means Global Accelerator.

Streaming and analytics services

SERVICE	WHAT IT DOES	SIGNAL
Kinesis Data Streams	Ordered, replayable stream	Multiple consumers, replay
Kinesis Data Firehose	Managed delivery to S3 or Redshift	'Simplest', near-real-time
Amazon MSK	Managed Apache Kafka	Kafka named explicitly
Amazon Athena	SQL over S3, pay per scan	Ad-hoc queries on S3 data
AWS Glue	Serverless ETL and catalogue	Transform without servers
Amazon EMR	Managed Spark and Hadoop	Heavy transformation
Amazon Redshift	Data warehouse	Frequent complex analytics

ON THE EXAM

Firehose delivers with no code. Data Streams lets you process and replay.

Moving data into AWS

SERVICE	CHOOSE IT WHEN
AWS DataSync	Ongoing or one-off transfer over the network, on-premises to AWS storage
AWS Snowball Edge	Tens of terabytes to petabytes, where the network would take too long
AWS Transfer Family	Existing partners use SFTP, FTPS or FTP and cannot change
Storage Gateway	On-premises applications need ongoing local access to cloud storage
S3 Transfer Acceleration	Uploads to S3 from distant geographies, over the internet
Database Migration Service	Migrating databases, including between different engines

ON THE EXAM

Do the arithmetic. If the transfer would take weeks over the available line, the answer is Snowball.

API Gateway, and when it appears

- ◆ **Fronts Lambda or any HTTP backend**, handling authorisation, throttling and versioning.
- ◆ **Throttling and usage plans** protect backends and support per-customer rate limits.
- ◆ **Caching responses** at the API layer reduces backend load for repeated requests.
- ◆ **Edge-optimised endpoints** route via CloudFront. Regional endpoints suit callers in one Region.
- ◆ **Direct service integrations** let API Gateway write straight to SQS or DynamoDB with no compute.

ON THE EXAM

'Rate limit different customers differently' is an API Gateway usage plan, not application code.

A media company serves large video files and images from an S3 bucket to users in forty countries. Users outside the bucket's Region report slow downloads, and the S3 data transfer bill is substantial.

Security requires that the bucket must not be publicly accessible, and content must be served only through the company's own domain.

Which solution addresses all of these?

- ☐ A Enable S3 Transfer Acceleration on the bucket
- ☐ B Create a CloudFront distribution with the bucket as origin, using Origin Access Control
- ☐ C Replicate the bucket to Regions in each continent and use latency-based routing
- ☐ D Make the bucket public and enable static website hosting

◆ PAUSE AND DECIDE

Which solution addresses all of these?

A Transfer Acceleration

Speeds uploads into S3 from distant places. This is a download problem, and it does nothing for the bucket's public access or cost.

C Replicate to many Regions

Would improve latency at the cost of storing several copies and managing replication. Far more expensive than caching.

B CloudFront with Origin Access Control

Caches near users to fix latency, cuts origin transfer cost, keeps the bucket private, and serves through a custom domain.

D Public bucket with website hosting

Directly violates the requirement that the bucket must not be publicly accessible.

THE SIGNAL

Transfer Acceleration is for uploads. CloudFront is for downloads. Mixing them up is a very common error.

A logistics company must move eighty terabytes of historical archives from its data centre to S3. The site has a 100 Mbps internet connection that is also used for daily operations and cannot be saturated.

The migration should complete within a month.

Which approach should be used?

- ☐ A AWS DataSync over the existing internet connection
- ☐ B AWS Snowball Edge devices
- ☐ C S3 Transfer Acceleration with multipart upload
- ☐ D Order an AWS Direct Connect circuit and transfer over it

◆ PAUSE AND DECIDE

Which approach should be used?

A DataSync over the internet

Eighty terabytes over 100 Mbps takes over ten weeks at full line rate, and the line cannot even be fully used.

C Transfer Acceleration

Optimises use of available bandwidth. It cannot create bandwidth that does not exist.

B Snowball Edge

Physical transfer sidesteps the bandwidth constraint entirely, and shipping and loading fit comfortably inside a month.

D Order Direct Connect

Provisioning typically takes weeks to months, so the circuit may not exist before the deadline, and it is a large expense for a one-off move.

THE SIGNAL

Do the division. Terabytes against line speed against deadline. That arithmetic is the entire question.

Clickstream events arrive at around fifty thousand records per second. They must be written to S3 in Parquet format for later analysis, with a delay of a minute or two being acceptable.

The team wants the simplest possible solution and does not want to write or operate any stream-processing code.

Which service should be used?

- ☐ A Kinesis Data Streams with a Lambda consumer writing to S3
- ☐ B Kinesis Data Firehose delivering to S3 with format conversion
- ☐ C Amazon MSK with a Kafka Connect S3 sink
- ☐ D SQS with a Lambda function batching writes to S3

◆ PAUSE AND DECIDE

Which service should be used?

A Data Streams with Lambda

Works, but you now own consumer code, batching, retries and shard management. The question rules that out.

C MSK with Kafka Connect

Powerful and the most operational work of any option. Kafka is not mentioned as an existing requirement.

B Firehose with format conversion

Fully managed delivery, converts to Parquet natively, buffers to hit the near-real-time window, and requires no code.

D SQS with Lambda

SQS distributes work rather than carrying an ordered high-throughput stream, and you would still write the batching code.

THE SIGNAL

'A minute or two is acceptable' plus 'no code' is the Firehose signature. Real-time with custom processing flips it to Data Streams.

A public REST API built on API Gateway and Lambda is used by two hundred partner organisations. One partner's faulty integration recently sent forty times its normal request volume, exhausting Lambda concurrency and degrading the API for everyone.

The company wants per-partner request limits and wants to avoid changing application code.

Which solution meets this requirement?

- ☐ A API Gateway usage plans with API keys per partner
- ☐ B AWS WAF rate-based rules on the API
- ☐ C Reserved concurrency on the Lambda function
- ☐ D Rate-limiting logic added to the Lambda function

◆ PAUSE AND DECIDE

Which solution meets this requirement?

- A** Usage plans with API keys
Purpose-built for per-consumer throttling. Each partner gets a key with its own rate and burst limits, configured not coded.
- B** WAF rate-based rules
Limits by source IP address rather than by partner identity. A partner behind several addresses, or sharing one, breaks the mapping.
- C** Reserved concurrency
Caps total function concurrency, which protects downstream systems but throttles all partners equally rather than the offender.
- D** Code-based rate limiting
Excluded by the requirement to avoid code changes, and it means the request has already consumed Lambda capacity before being rejected.

THE SIGNAL

Per-consumer limits mean API Gateway usage plans. Reserved concurrency protects the backend but cannot distinguish between callers.

A hospital runs an imaging application on premises that must read and write to a local file share for performance reasons, but the organisation wants the data durably stored in S3 and available to cloud analytics.

The local site has limited storage capacity and cannot hold the full archive, but staff must be able to open recent studies immediately.

Which solution meets these requirements?

- ☐ A AWS Storage Gateway in File Gateway mode
- ☐ B AWS DataSync on an hourly schedule
- ☐ C Amazon EFS mounted over a VPN from on premises
- ☐ D AWS Snowball Edge kept permanently on site

◆ PAUSE AND DECIDE

Which solution meets these requirements?

- A** File Gateway
Presents a local file share backed by S3, caching recently used data locally while the full archive lives in the cloud.
- B** DataSync hourly
Copies data on a schedule. It does not present a local share, and the site cannot hold the full archive anyway.
- C** EFS over VPN
Possible but every read crosses the network, which fails the local performance requirement.
- D** Snowball on site
Snowball is a transfer device, not permanent hybrid storage. It offers no ongoing S3 integration.

THE SIGNAL

'On-premises application needs a local file share, but the data lives in S3' is always Storage Gateway.

Three hundred external suppliers upload nightly inventory files to the company's servers using SFTP. Their systems are old and the company cannot ask them to change how they connect.

The company wants the files to land in S3 for processing, and wants to stop maintaining its own SFTP servers.

Which solution should be used?

- ☐ A AWS Transfer Family with an SFTP endpoint backed by S3
- ☐ B Provide each supplier with IAM credentials and the AWS CLI
- ☐ C Run SFTP servers on EC2 in an Auto Scaling group
- ☐ D Issue presigned S3 URLs to each supplier nightly

◆ PAUSE AND DECIDE

Which solution should be used?

- A** AWS Transfer Family
A managed SFTP endpoint that writes straight into S3. Suppliers connect exactly as before, and there are no servers to maintain.
- B** IAM credentials and the CLI
Requires all three hundred suppliers to change how they connect, which the question forbids.
- C** SFTP on EC2
Keeps the servers the company wants to stop maintaining, merely moving them to AWS.
- D** Presigned URLs nightly
Changes the upload mechanism, and generating and distributing URLs to three hundred suppliers every night is significant new work.

THE SIGNAL

Transfer Family has one signal: external parties using SFTP, FTPS or FTP who cannot change.

A retail analytics platform stores five years of transaction data in S3, partitioned by date. Business analysts run around four hundred complex queries per day involving joins across several large tables, and expect dashboards to respond in under two seconds.

The team currently uses Athena and reports that dashboards are too slow and that costs have risen sharply.

What should the solutions architect recommend?

- ☐ A Keep Athena and add more partitions to reduce data scanned
- ☐ B Load the data into Amazon Redshift and point the dashboards at it
- ☐ C Move the data into DynamoDB with a global secondary index
- ☐ D Run the queries on an EMR cluster with Spark SQL

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 More partitions in Athena
Genuinely reduces cost per query, but Athena is not built for sub-second dashboards, and four hundred queries a day makes pay-per-scan expensive.
- B

 Load into Redshift
A warehouse built for frequent complex joins with fast, predictable response. At this query volume a running cluster costs less than per-scan pricing.
- C

 DynamoDB with a GSI
A key-value store. Joins across several large tables are precisely what it cannot do.
- D

 EMR with Spark SQL
Suited to heavy transformation rather than interactive dashboards, and it carries the most operational overhead.

THE SIGNAL

This is the deliberate inverse of the Athena scenario in module eight. Frequency and complexity move the answer from Athena to Redshift.

A company is migrating a 4 TB Oracle database to Amazon Aurora PostgreSQL. The application must continue running throughout, with a cutover window of no more than fifteen minutes.

The schemas use Oracle-specific data types and stored procedures that have no direct PostgreSQL equivalent.

Which combination of services should be used?

- ☐ A AWS Database Migration Service with the Schema Conversion Tool, using ongoing replication
- ☐ B Export to S3 with Snowball and import into Aurora
- ☐ C Take an Oracle dump, restore it into Aurora, and switch over
- ☐ D AWS DataSync to copy the database files to Aurora storage

◆ PAUSE AND DECIDE

Which combination of services should be used?

- A** DMS with SCT and ongoing replication
SCT converts the incompatible schema and procedures; DMS does a full load then replicates changes continuously, so cutover is only the final catch-up.
- B** Snowball export and import
Snowball addresses bandwidth, which is not the problem at 4 TB, and it gives no mechanism for continuous replication.
- C** Dump and restore
Requires downtime for the whole load, and a native Oracle dump does not convert Oracle-specific types into PostgreSQL.
- D** DataSync to Aurora storage
DataSync moves files between file and object storage. You cannot write database files into Aurora's managed storage.

THE SIGNAL

Different engines means SCT. Minimal downtime means DMS ongoing replication. The two together are the standard heterogeneous migration answer.

Module 9 takeaways

- ◆ CloudFront for downloads, Transfer Acceleration for uploads. Do not swap them.
- ◆ Firehose when a minute is fine and no code is wanted. Data Streams when you process records yourself.
- ◆ Do the bandwidth arithmetic. If it takes longer than the deadline, the answer is Snowball.
- ◆ DataSync transfers. Storage Gateway provides ongoing local access.
- ◆ Athena for infrequent ad-hoc. Redshift for frequent complex analytics.

ON THE EXAM

That completes domain three. Next: the three cost modules, twenty percent of the exam and the easiest marks to win.